Xiaoyu Wang

Madison, WI | (585) 851-3818 | xwang2696@wisc.edu
Personal Website | LinkedIn | GitHub | Google Scholar

EDUCATION

Ph.D. in Biological Systems Engineering *Jan 2023 – Present*
University of Wisconsin–Madison, Madison, WI
Minor: Computer Sciences
Relevant Coursework: CS 533 (Image Processing); CS 760 (Machine Learning); CS 759 (High Performance Computing for Applications in Engineering)

Doctoral Studies in Computing and Information Sciences *Aug 2022 – 2023*
Rochester Institute of Technology, Rochester, NY
Golisano College of Computing and Information Sciences
Relevant Coursework: CSCI 820 (Quantitative Foundations); CSCI 865 (Deep Learning)

B.S. in Computer Science and Technology *2017 – 2021*
Xi'an Jiaotong University, Xi'an, China

EXPERIENCE

Graduate Research Assistant *Jan 2023 – Present*
University of Wisconsin–Madison, Madison, WI
Supervised by: Prof. Zhou Zhang

- Developed and deployed a full-stack cyber-agricultural web platform using Vue 3, MapLibre GL JS, FastAPI, and Vite to publish interactive county-level crop-yield predictions.
- Built geospatial visualization and backend API workflows that deliver in-season county estimates ahead of USDA NASS official releases for researchers, policymakers, analysts, and agricultural stakeholders.
- Structured a frontend/backend monorepo and production workflow with Vite proxying, Gunicorn serving, and deployment automation; launched the live Web Tool and released the source code.

Research Intern *Apr 2021 – Apr 2022*
Microsoft Research Asia, Media Computing Group, Beijing, China
Supervised by: Xiangyu Kong and Xiulian Peng

- Designed a multi-modal multi-correlation learning framework for audio-visual speech separation that explicitly modeled speaker-identity and phonetic correlations using contrastive and adversarial training.
- Integrated a U-Net separation backbone with wav2vec 2.0, lip-reading, and ResNet-18 speaker-identification models in PyTorch; trained and evaluated models on four NVIDIA V100 GPUs.
- Improved LRS3 SDR from 8.46 to 9.98, PESQ from 2.27 to 2.584, and STOI from 0.843 to 0.861 without increasing inference complexity; published a first-author paper at Interspeech 2022.

Research Intern *Sep 2020 – Mar 2021*
SenseTime, Autonomous Driving Group
Supervised by: Guohang Yan and Tao Ma

- Contributed to OpenCalib, an open-source multi-sensor calibration toolbox for autonomous driving, by implementing calibration workflows among LiDAR, radar, and camera sensors.
- Trained a deep learning model to determine whether multi-sensor calibration results were accurate, automating calibration-quality assessment.

Remote Research Intern *Jun 2020 – Sep 2020*
Nanyang Technological University
Supervised by: Tao Bai and Jun Zhao

- Proposed a data-free targeted universal adversarial perturbation method with a custom loss that balanced model disruption and target-label objectives without access to training data.
- Extended the attack to federated-learning settings and published the work as a first-author paper at SciSec 2021.

Research Intern *Jun 2019 – Feb 2020*
INNNO

- Built a remote-sensing change-detection pipeline for drone imagery, including image alignment, bi-temporal image preprocessing, and semantic-segmentation mask prediction.
- Created synthetic change pairs to address limited labeled data and trained a Python-based model to detect changes in buildings, waterways, and vegetation; released the project code.

Research Assistant *Jan 2019 – Sep 2019*
College of Artificial Intelligence, Xi'an Jiaotong University
Supervised by: Jinjun Wang

- Developed Distributed-GAN, enabling multiple users to train generative models locally on private data without sharing raw datasets while improving generated-sample diversity.
- Published the work as a first-author arXiv preprint.

Research Assistant *Oct 2017 – 2018*
Xi'an Jiaotong University
Supervised by: Jinsong Han
Conducted research on computer security and deep learning.

PUBLICATIONS

Journal Articles

Assimilation of LAI and FAPAR within DSSAT Guided by Leaf Growth Dynamics to Enhance Maize Yield Phenotyping 2026
*Zezhong Tian, Jiahao Fan, Tong Yu, **Xiaoyu Wang**, Mingda Wu, Natalia de Leon, Shawn M. Kaeppler, Zhou Zhang*
Computers and Electronics in Agriculture, vol. 250, article 111946, 2026.

Learning County from Pixels: Corn Yield Prediction with Attention-Weighted Multiple Instance Learning 2025
***Xiaoyu Wang**, Yuchi Ma, Yijia Xu, Qunying Huang, Zhengwei Yang, Zhou Zhang*
International Journal of Remote Sensing, vol. 46, no. 7, pp. 2815–2845, 2025.

Book Chapter

Satellite-Based Mapping of Crop Yield 2026
*Yuchi Ma, Shuo Chen, **Xiaoyu Wang**, Zhou Zhang, Zhengwei Yang, Qunying Huang, Xinghua Cheng, Gengchen Mai*
In Agricultural Applications of Earth Observation, ch. 7, pp. 193–230, Elsevier, 2026.

Conference Papers

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025
*Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang*
Accepted to IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2025.

County-Level Crop Yield Prediction Using SMAP-Derived Data Products and Deep Learning Model 2024
*Zhengwei Yang, **Xiaoyu Wang**, Jingyi Huang, Zhou Zhang*
IEEE International Geoscience and Remote Sensing Symposium (IGARSS), pp. 3222–3225, 2024.

Multi-Modal Multi-Correlation Learning for Audio-Visual Speech Separation 2022
***Xiaoyu Wang**, Xiangyu Kong, Xiulian Peng, Yan Lu*
Interspeech, 2022.

[*A Data-Free Approach for Targeted Universal Adversarial Perturbation*] 2021
Xiaoyu Wang, *Tao Bai, Jun Zhao*
Science of Cyber Security (SciSec), pp. 126–138, 2021.

Preprint

Knowledge-Guided Machine Learning Model with Soil Moisture for Corn Yield Prediction under Drought Conditions 2025
Xiaoyu Wang, *Yijia Xu, Jingyi Huang, Zhengwei Yang, Zhou Zhang*
arXiv:2503.16328, 2025.

AWARDS & HONORS

Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia, 2021.
BSE Travel Award, University of Wisconsin–Madison, 2025. ($1,000)
Third-Class Award, Xi'an Jiaotong University, 2017. (Top 20% by GPA)

PROFESSIONAL SERVICE

Journal Reviewer
International Journal of Applied Earth Observation and Geoinformation (JAG)

TECHNICAL SKILLS

Languages:	Python, C++, Shell, MATLAB, CUDA C
Machine Learning & GPU:	PyTorch, TensorFlow, TensorRT, CUDA, cuDNN, cuBLAS
Web & Deployment:	FastAPI, Vue 3, MapLibre GL JS, Vite, Gunicorn, Docker
GIS & Remote Sensing:	Google Earth Engine, GDAL, QGIS, ArcGIS
Vision, Audio & Robotics:	OpenCV, librosa, Asteroid, Kaldi, FFmpeg, PCL, ROS
Development:	Linux, Git, CMake